



Appendix B2

TRAINING SYLLABUS FOR PPL theory

**Intended for training of pilots
into the EASA PPL(A)
theoretical knowledge**

SUMMARY:

Subject		No of	Hours
010 ALW	Air Law	4	12
021 AGK	Aircraft General Knowledge	2	6
022 AGK Instr	Aircraft General Knowledge Instrumentation	1	4
031 M&B	Mass and Balance	1	4
032 PER	Flight Performance	2	8
033 FPL	Flight Planning	2	8
040 HPL	Human Performance and Limitations	2	6
050 MET	Meteorology	4	16
061 NAV Gen	General Navigation	2	8
062 NAV Radic	Radio Navigation	2	6
070 OPP	Operational Procedures	2	6
080 POF	Principals of Flight	4	13
091 COM VFR	Communication VFR	1	3



PPL Training Manual

2025-03-05

Rev 4

App B2

AIR LAW AND ATC PROCEDURES – Lesson 1

Covered LO:

010 00 00 00 AIR LAW AND ATC PROCEDURES
010 01 00 00 INTERNATIONAL LAW: CONVENTIONS, AGREEMENTS AND ORGANISATIONS
010 02 00 00 ANNEX 8 – AIRWORTHINESS OF AIRCRAFT
010 03 00 00 ANNEX 7 – AIRCRAFT NATIONALITY AND REGISTRATION MARKS
010 04 00 00 ANNEX 1 – PERSONNEL LICENSING

Teaching materials:

1. SSFA LO PPL(A) 010 Air Law
2. ICAO Annexes 8, 7, 1.
3. KSAB PPL, Luftfartssystemet och Säkerhetsstandarder
4. Oxford or Jeppesen “Air Law”, chapters included in PPL(A) LO requirements.
5. SSFA Lecture presentations 010 Air Law

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

INTERNATIONAL LAW: CONVENTIONS, AGREEMENTS AND ORGANISATIONS

- The Convention on international civil aviation (Chicago) Doc. 7300/6
- Part I Air Navigation – relevant parts of the following chapters: general principles and application of the convention
- Flight over territory of Contracting States nationality of aircraft measures to facilitate air navigation conditions to be fulfilled with respect to aircraft.
- International standards and recommended practices validity of endorsed certificates and licenses notification of differences
- International Civil Aviation Organization (ICAO) – objectives and composition

ANNEX 8 – AIRWORTHINESS OF AIRCRAFT

- Foreword, definitions
- Certificate of airworthiness

ANNEX 7 – AIRCRAFT NATIONALITY AND REGISTRATION MARKS

- Foreword, definitions
- Common and registration marks
- Certificate of registration, Aircraft nationality

ANNEX 1 – PERSONNEL LICENSING

- Definitions
- Relevant parts of Annex 1 connected to PART-FCL and PART-MED

TEM topics:

- Safety procedures, regulated vs nonregulated

Progress testing:

PT ALW Lesson 01 performed at lessons end.

Interconnection:

- Legal theory vs practice
- Local variations in interpretation/execution

Mandatory student preparation:

- Flygcert ALW CBT chapters 1-3



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AIR LAW AND ATC PROCEDURES – Lesson 1

Learning Objectives

010 00 00 00 AIR LAW AND ATC PROCEDURES

010 01 00 00 INTERNATIONAL LAW: CONVENTIONS, AGREEMENTS AND ORGANISATIONS

010 01 01 00 The Convention on international civil aviation (Chicago) Doc. 7300/6

010 01 01 01 Part I Air Navigation – relevant parts of the following chapters:

- general principles and application of the convention
 - flight over territory of Contracting States nationality of aircraft
 - measures to facilitate air navigation conditions to be fulfilled with respect to aircraft
 - international standards and recommended practices validity of endorsed certificates and licenses
- notification of differences

010 01 01 02 Part II the International Civil Aviation Organization (ICAO) – objectives and composition

010 02 00 00 ANNEX 8 – AIRWORTHINESS OF AIRCRAFT

010 02 01 00 Foreword, definitions

010 02 02 00 Certificate of airworthiness

010 03 00 00 ANNEX 7 – AIRCRAFT NATIONALITY AND REGISTRATION MARKS

010 03 01 00 Foreword, definitions

010 03 02 00 Common and registration marks

010 03 03 00 Certificate of registration, Aircraft nationality

010 04 00 00 ANNEX 1 – PERSONNEL LICENSING

010 04 01 01 Definitions

010 04 02 00 Relevant parts of Annex 1 connected to PART-FCL and PART-MED

AIR LAW AND ATC PROCEDURES – Lesson 2

Overview

010 05 00 00 RULES OF THE AIR

010 06 00 00 PROCEDURES FOR AIR NAVIGATION – AIRCRAFT OPERATIONS Doc. 8168
PANS-OPS

Ref.:

1. SSFA LO PPL(A) 010 Air Law
2. ICAO Annex 2
3. ICAO Doc 8168 PANS-OPS
4. KSAB PPL, Luftfartssystemet och Säkerhetsstandarder
5. Oxford or Jeppesen “Air Law”, chapters included in PPL(A) LO requirements.
6. SSFA Lecture presentations 010 Air Law

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

ANNEX 2 RULES OF THE AIR

- Essential definitions, applicability of the rules of the air, general rules (except water operations).
- Describe the general rules and regulations that deal with minimum flight altitude, rules that apply to avoid collision and the lights to be carried by aircraft.
- Describe VFR Flight Rules
- Describe signals related to pilot.
- Describe rules for intercepting civil aircraft.

PROCEDURES FOR AIR NAVIGATION

- Altimeter setting procedures.
- Basic requirements (except tables), procedures applicable to operators and pilots (except tables)
- Secondary surveillance radar transponder operating procedures (including ICAO Doc. 7030 regional supplementary procedures)
- Operation of transponders
- Phraseology

TEM topics:

- See and be seen

Progress testing:

PT ALW Lesson 02 performed at lessons end.

Interconnection:

- Emergency procedures
- Separation, control, alerting Services, visual Separation
- Procedures for Flight Control Services
- Radar services

Mandatory student preparation:

- Flygcert ALW CBT chapters 4-6

AIR LAW AND ATC PROCEDURES – Lesson 2

Learning Objectives

010 05 00 00 ANNEX 2 RULES OF THE AIR

010 05 01 00 Essential definitions, applicability of the rules of the air, general rules (except water operations)

010 05 01 03 Describe the general rules and regulations that deal with minimum flight altitude, rules that apply to avoid collision and the lights to be carried by aircraft.

010 05 01 06 Describe VFR Flight Rules

010 05 01 07 Describe signals related to pilot.

010 05 01 08 Describe rules for intercepting civil aircraft.

010 06 00 00 PROCEDURES FOR AIR NAVIGATION – AIRCRAFT OPERATIONS Doc. 8168 OPS/611, VOLUME 1

010 06 06 00 Altimeter setting procedures.

010 06 06 01 Basic requirements (except tables), procedures applicable to operators and pilots (except tables)

010 06 08 00 Secondary surveillance radar transponder operating procedures (including ICAO Doc. 7030

– regional supplementary procedures)

010 06 08 01 Operation of transponders

010 06 08 02 Phraseology

AIR LAW AND ATC PROCEDURES – Lesson 3

Overview

010 07 00 00 AIR TRAFFIC SERVICES AND AIR TRAFFIC MANAGEMENT
010 08 00 00 AERONAUTICAL INFORMATION SERVICE

Ref.:

1. SSFA LO PPL(A) 010 Air Law
2. ICAO Annex 11
3. ICAO Doc 4444
4. ICAO Annex 15
5. KSAB PPL, Luftfartssystemet och Säkerhetsstandarder
6. Oxford or Jeppesen “Air Law”, chapters included in PPL(A) LO requirements.
7. SSFA Lecture presentations 010 Air Law

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

AIR TRAFFIC MANAGEMENT SERVICE (ATS) AND AIR MANAGEMENT SERVICE (ATM)

- Annex 11 – Air Traffic Services
- Definition, Explain the contents of Annex 11 to the Chicago Convention regarding essential concepts, classification, tasks and distribution of responsibilities between different ATS units.
- Describe the following areas within which air traffic management services are provided: FIR, CTA, TMA, CTR, ATZ, TIA, TIZ, and briefly describe the tasks of the air traffic management service in general and its division.
- Air traffic control service ATC
- Aerodrome Control Service TWR
 - Procedures for aerodrome control services and responsibilities
 - Traffic Pattern
 - Visual separation in the vicinity of aerodromes.
 - Describe safety related terms such as braking action and wake turbulence.
- Flight Information Services (FIS)
- Alerting Services
- Visual Separation
- Procedures for Flight Control Services
- Radar services
- Procedures related to emergencies, communication failure and contingencies.

ANNEX 15 AERONAUTICAL INFORMATION SERVICE

- Introduction,
- Essential Definitions
- AIP, NOTAM, AIRAC, AIC

TEM topics:

- Clearances, confirming receipt and/or “say again”.

Progress testing:

PT ALW Lesson 03 performed at lessons end.

Interconnection:

- Emergency procedures
- Separation, control, alerting Services, visual Separation

Mandatory student preparation:

- Flygcert ALW CBT chapters 7-9

AIR LAW AND ATC PROCEDURES – Lesson 3

Learning Objectives

010 07 00 00 Air Traffic Management Service (ATS) and Air Management Service (ATM)
010 07 01 00 Annex 11 – Air Traffic Services
010 07 01 01 Definition, Explain the contents of Annex 11 to the Chicago Convention regarding essential concepts, classification, tasks and distribution of responsibilities between different ATS units.
010 07 01 02 Describe the following areas within which air traffic management services are provided: FIR, CTA, TMA, CTR, ATZ, TIA, TIZ, and briefly describe the tasks of the air traffic management service in general and its division.
010 07 01 03 Air traffic control service ATC
010 07 01 04 Aerodrome Control Service TWR
- Procedures for aerodrome control services and responsibilities
- Traffic Pattern
- Visual separation in the vicinity of aerodromes.
- Describe safety related terms such as braking action and wake turbulence.
010 07 01 05 Flight Information Services (FIS)
010 07 01 06 Alerting Services
010 07 02 13 Visual Separation
010 07 02 16 Procedures for Flight Control Services
010 07 02 17 Radar services
010 07 02 19 Procedures related to emergencies, communication failure and contingencies.
010 08 00 00 ANNEX 15 AERONAUTICAL INFORMATION SERVICE
010 08 01 00 Introduction,
010 08 02 00 Essential Definitions
010 08 03 00 AIP, NOTAM, AIRAC, AIC

AIR LAW AND ATC PROCEDURES – Lesson 4

Overview

010 09 00 00 AERODROMES (ICAO Annex 14, Volume I, Aerodrome Design and Operations)
010 11 00 00 ANNEX 12 SEARCH AND RESCUE
010 12 00 00 ANNEX 17 SECURITY
010 13 00 00 ANNEX 13 AIRCRAFT ACCIDENT INVESTIGATION
010 15 00 00 NATIONAL LAW

Repetition
School Test
Preparation for Exam

Ref.:

1. SSFA LO PPL(A) 010 Air Law
2. ICAO Annexes 12, 13, 14, 17
3. KSAB PPL, Luftfartssystemet och Säkerhetsstandarder
4. Oxford or Jeppesen “Air Law”, chapters included in PPL(A) LO requirements.
5. SSFA Lecture presentations 010 Air Law

Setting:
Classroom

Time allocated:
Instructor led study: 3 hrs.

Lesson content

ANNEX 14, Vol 1 & 2 AERODROMES

- Essential definitions

Aerodrome data:

- Conditions of the movement area and related facilities
- Visual aids for denoting obstacles
- Wind direction indicator, wind cone
- Way marking of obstacles
- Unpaved taxiway
- Unserviceable surfaces
- Snow-covered surfaces
- Fixed obstacles
- Temporary obstacles
- Moving obstacles.
- Wind direction Indicator and requirements

Markings

- Track number marking, including the meaning of the track numbers
- Taxiway
- Plate
- Plate, safety marking
- Threshold marking (normal and moved-in threshold)
- Marking of unpaved lane
- Marking of paved track
- Marking of waiting area
- Marking of non-load-bearing surfaces

Lights

- Colours and functions
- Bench light
- Threshold light
- Tunway end lights
- Taxi edge lights
- Lane warning light in use.

Signs

- Visual aids to mark obstacles.
- Visual aids to mark obstacles.

- Marking of objects
- Obstacle lighting
- Visual aids for marking of areas with limited usage.
- Operational airport regulations

ANNEX 12 SEARCH AND RESCUE

- Essential definitions
- Operating procedures
- Procedures for pilots in command at the scene of an accident
- Procedures for pilots in command intercepting a distress transmission
- Search and rescue signals

ANNEX 17 SECURITY

- General: aims and objectives

ANNEX 13 AIRCRAFT ACCIDENT INVESTIGATION

- Essential definitions
- Applicability

TEM topics:

- Movement areas and airport markings

Progress testing:

PT ALW Lesson 04 performed at lessons end.

Interconnection:

- Take off, land, taxi and park

Mandatory student preparation:

- Flygcert ALW CBT chapters 12-13

AIR LAW AND ATC PROCEDURES – Lesson 4

Learning Objectives

- 010 09 00 00 ANNEX 14, Vol 1 & 2 AERODROMES
- 010 09 01 00 Essential definitions
- 010 09 02 00 Aerodrome data:
 - conditions of the movement area and related facilities
- 010 09 04 00 Visual aids for denoting obstacles
 - wind direction indicator, wind cone
 - day marking of obstacles
 - unpaved taxiway
 - unserviceable surfaces
 - snow-covered surfaces
 - fixed obstacles
 - temporary obstacles
 - moving obstacles.
- 010 09 04 01 Wind direction Indicator and requirements
- 010 09 04 02 Markings
 - track number marking, including the meaning of the track numbers
 - roller coaster
 - taxiway
 - plate
 - plate, safety marking
 - threshold marking (normal and moved-in threshold)
 - marking of unpaved lane
 - marking of paved track
 - marking of waiting area
 - marking of non-load-bearing surfaces
- 010 09 04 03 Lights, colours and functions
 - bench light
 - threshold light
 - runway end lights
 - taxi edge lights
 - lane warning light in use.
- 010 09 04 04 Signs
- 010 09 04 05 Visual aids to mark obstacles.
- 010 09 05 00 Visual aids to mark obstacles.
- 010 09 05 01 Marking of objects
- 010 09 05 01 Obstacle lighting
- 010 09 06 00 Visual aids for marking of areas with limited usage.
- 010 09 07 00 Operational airport regulations
- 010 11 00 00 ANNEX 12 SEARCH AND RESCUE
- 010 11 01 00 Essential definitions
- 010 11 03 00 Operating procedures
 - procedures for pilots in command at the scene of an accident
 - procedures for pilots in command intercepting a distress transmission
 - search and rescue signals
- 010 12 00 00 ANNEX 17 SECURITY
 - General: aims and objectives
- 010 13 00 00 ANNEX 13 AIRCRAFT ACCIDENT INVESTIGATION
- 010 13 01 00 Essential definitions
- 010 13 02 00 Applicability

Aircraft General Knowledge – Lesson 1

Overview

021 00 00 00 AIRCRAFT GENERAL KNOWLEDGE – AIRFRAME AND SYSTEMS, ELECTRICS, POWERPLANT, EMERGENCY EQUIPMENT
021 01 00 00 SYSTEM DESIGN, LOADS, STRESSES, MAINTENANCE
021 02 00 00 AIRFRAME
021 03 00 00 HYDRAULICS
021 04 00 00 LANDING GEAR, WHEELS, TYRES, BRAKES
021 05 00 00 FLIGHT CONTROLS
021 08 00 00 Fuel System

Ref.:

1. SSFA LO PPL(A) 021 Aircraft General Knowledge
2. KSAB PPL Luftfartyg generellt
3. Oxford or Jeppesen “Airframes”, chapters included in PPL(A) LO requirements.
4. Oxford or Jeppesen “Systems”, chapters included in PPL(A) LO requirements.
5. Oxford or Jeppesen “Electrics”, chapters included in PPL(A) LO requirements.
6. SSFA Lecture presentations 021 Aircraft General Knowledge

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

AIRCRAFT GENERAL KNOWLEDGE – AIRFRAME AND SYSTEMS, ELECTRICS, POWERPLANT, EMERGENCY EQUIPMENT SYSTEM DESIGN, LOADS, STRESSES, MAINTENANCE

System Design

- Design concept (safe life, Fail-safe, damage tolerant)
- Loads and combination loadings applied to an aircraft 's structure.
- Aircraft Maintenance aircraft maintenance method: hard time and on-condition.

AIRFRAME

- Construction and joining method.
- Describe monocoque, semi-monocoque and sandwich structures.
- Describe the following connection methods: riveting, soldering, bolted joints, bonding.
- Describe properties of Aluminum, steel and composite materials.
- Wings, tail surfaces and control surfaces
- Describe different type of wing like cantilever and braced.
- Describe structural component of a wing
- Fuselage, doors, floor, wind screen and windows

HYDRAULICS

- Hydromechanics: basic principles
- Hydraulic systems
- Describe the components of a simple hydraulic system with regard to: construction and features.

LANDING GEAR, WHEELS, TYRES, BRAKES

- Landing gear
- Nose wheel steering: design, function
- Brakes
- Wheels and Tires
- FLIGHT CONTROLS
- Primary Flight Controls
- Secondary Flight Controls
- De-icing system
- Fuel System
- Piston engine

FUEL SYSTEM

- Describe functions of fuel system components
- Describe difference between gravity-feed and pressure-feed system.

- Describe different fuel tank: integral, built-in and rubber tank.

TEM topics:

- System failures, mechanical failures, electrical failures.

Progress testing:

PT AGK Lesson 01 performed at lessons end.

Interconnection:

- Recognition of system failures, recovery/emergency procedures

Mandatory student preparation:

- Flygcert AGK CBT chapters 1-6

Aircraft General Knowledge – Lesson 1

Learning Objectives

021 00 00 00 AIRCRAFT GENERAL KNOWLEDGE – AIRFRAME AND SYSTEMS, ELECTRICS, POWERPLANT, EMERGENCY EQUIPMENT

021 01 00 00 SYSTEM DESIGN, LOADS, STRESSES, MAINTENANCE

021 01 01 00 System Design

021 01 01 01 Design concept (safe life, Fail-safe, damage tolerant)

021 01 02 00 Loads and combination loadings applied to an aircraft 's structure.

021 01 05 00 Aircraft Maintenance

021 01 05 01 aircraft maintenance method: hard time and on-condition.

021 02 00 00 AIRFRAME

021 02 01 00 Construction and joining method.

- Describe monocoque, semi-monocoque and sandwich structures.
- Describe the following connection methods: riveting, soldering, bolted joints, bonding.
- Describe properties of Aluminum, steel and composite materials.

021 02 03 00 Wings, tail surfaces and control surfaces

- Describe different type of wing like cantilever and braced.
- Describe structural component of a wing

021 02 04 00 Fuselage, doors, floor, wind screen and windows

021 03 00 00 HYDRAULICS

021 03 01 00 Hydromechanics: basic principles

021 03 02 00 Hydraulic systems

Describe the components of a simple hydraulic system with regard to: construction and features.

021 04 00 00 LANDING GEAR, WHEELS, TYRES, BRAKES

021 04 01 00 Landing gear

021 04 02 00 Nose wheel steering: design, function

021 04 03 00 Brakes

021 04 04 00 Wheels and Tires

021 05 00 00 FLIGHT CONTROLS

021 05 01 00 Primary Flight Controls

021 05 02 00 Secondary Flight Controls

021 05 03 00 De-icing system

021 08 00 00 Fuel System

021 08 01 00 Piston engine

- Explain fuel system task
- Describe functions of fuel system components
- Describe difference between gravity-feed and pressure-feed system.
- Describe different fuel tank: integral, built-in and rubber tank.

Aircraft General Knowledge – Lesson 2

Overview

021 09 00 00 ELECTRICS

021 10 00 00 PISTON ENGINES

Repetition

School Test

Preparation for Exam

Ref.:

1. SSFA LO PPL(A) 021 Aircraft General Knowledge
2. KSAB PPL Luftfartyg generellt
3. Oxford or Jeppesen “Airframes”, chapters included in PPL(A) LO requirements.
4. Oxford or Jeppesen “Systems”, chapters included in PPL(A) LO requirements.
5. Oxford or Jeppesen “Electrics”, chapters included in PPL(A) LO requirements.
6. SSFA Lecture presentations 021 Aircraft General Knowledge

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

ELECTRICS

- Electrics: general, definitions
- Static Electricity
- Direct Current: voltage, current, resistance, conductivity, Ohm 's law, power, work
- Alternating Current: voltage, current, amplitude, phase, frequency, resistance
- Circuits: series, parallel
- Magnetic field: effects in an electrical circuit

Batteries

- Types, characteristics, limitations and limitations

Power Generating

- AC and DC power generation, distribution and consumption
- Power Distribution
- Explain the function of power distribution.

PISTON ENGINES

- General
- Types of internal combustion engine: basic principles, definitions
- Fuel
- Types, grades, characteristics, limitations
- Carburetor/Injection system
- Carburetor: design, operation, degraded modes of operation, indications and warnings
- Injection: design, operation, degraded modes of operation, indications and warnings, icing

Air cooling systems

- Design, operation, degraded modes of operation, indications and warnings
- Lubrication systems
- Lubricants: types, characteristics, limitations
- Design, operation, degraded modes of operation, indications and warnings
- Ignition circuits
- Design, operation, degraded modes of operation
- Mixture
- Definition, characteristic mixtures, control instruments, associated control levers, indications

Propellers

- Definitions, general: Design, operation, system components, variations
- Reduction Gear, Construction
- Propeller handling: Associated control levers, indications and warnings

Performance and engine handling

- Influence of engine parameters, influence of atmospheric conditions, limitations, power augmentation systems
- Engine handling: power and mixture settings during various flight phases, operational limitations

TEM topics:

- Recognition of system failures, mechanical failures, electrical failures.

Progress testing:

PT AGK Lesson 02 performed at lessons end.

Interconnection:

- Recognition of/recovery from system failures.

Mandatory student preparation:

- Flygcert AGK CBT chapters 6-13

Aircraft General Knowledge – Lesson 2

Learning Objectives

021 09 00 00 ELECTRICS
021 09 01 00 Electrics: general, definitions
021 09 01 01 Static Electricity
021 09 01 02 Direct Current:
voltage, current, resistance, conductivity, Ohm 's law, power, work
021 09 01 03 Alternating Current:
voltage, current, amplitude, phase, frequency, resistance
021 09 01 04 Circuits: series, parallel
021 09 01 05 Magnetic field: effects in an electrical circuit
021 09 02 00 Batteries
021 09 02 01 Types, characteristics, limitations and limitations
021 09 03 00 Power Generating
021 09 03 01 AC and DC power generation, distribution and consumption
021 09 04 00 Power Distribution
021 09 04 01 Explain the function of power distribution.
021 10 00 00 PISTON ENGINES
021 10 01 00 General
021 10 01 01 Types of internal combustion engine: basic principles, definitions
021 10 02 00 Fuel
021 10 02 01 Types, grades, characteristics, limitations
021 10 04 00 Carburetor/Injection system
021 10 04 01 Carburetor: design, operation, degraded modes of operation, indications and warnings
021 10 04 02 Injection: design, operation, degraded modes of operation, indications and warnings
021 10 04 03 Icing
021 10 05 00 Air cooling systems
021 10 05 01 Design, operation, degraded modes of operation, indications and warnings
021 10 06 00 Lubrication systems
021 10 06 01 Lubricants: types, characteristics, limitations
021 10 06 02 Design, operation, degraded modes of operation, indications and warnings
021 10 07 00 Ignition circuits
021 10 07 01 Design, operation, degraded modes of operation
021 10 08 00 Mixture
021 10 08 01 Definition, characteristic mixtures, control instruments, associated control
levers, indications
021 10 09 00 Propellers
021 10 09 01 Definitions, general: Design, operation, system components, variations
021 10 09 03 Reduction Gear, Construction
021 10 09 04 Propeller handling: Associated control levers, indications and warnings
021 10 10 00 Performance and engine handling
021 10 10 01 Performance: influence of engine parameters, influence of atmospheric
conditions, limitations, power augmentation systems
021 10 10 02 Engine handling: power and mixture settings during various flight phases, operational
limitations

AIRCRAFT GENERAL KNOWLEDGE – INSTRUMENTATION – Lesson 1

Overview

022 00 00 00 INSTRUMENTATION
022 01 00 00 INSTRUMENT AND INDICATION SYSTEMS
022 02 00 00 MEASUREMENT OF AERODYNAMIC PARAMETERS
022 03 00 00 MAGNETISM – DIRECT READING COMPASS
022 04 00 00 GYROSCOPIC INSTRUMENTS
022 10 00 00 COMMUNICATION SYSTEMS
022 12 00 00 ALERTING SYSTEMS, PROXIMITY SYSTEMS
022 13 00 00 INTEGRATED INSTRUMENTS – ELECTRONIC DISPLAYS

Repetition
School Test
Preparation for Exam

Ref.:

1. SSFA LO PPL(A) 022 Aircraft General Knowledge
2. KSAB PPL Luftfartyg generellt
3. Oxford or Jeppesen “Airframes”, chapters included in PPL(A) LO requirements.
4. Oxford or Jeppesen “Systems”, chapters included in PPL(A) LO requirements.
5. Oxford or Jeppesen “Instruments”, chapters included in PPL(A) LO requirements.
6. Oxford or Jeppesen “Electrics”, chapters included in PPL(A) LO requirements.
7. SSFA Lecture presentations 022 Aircraft General Knowledge.

Setting:
Classroom

Time allocated:
Instructor led study: 4 hrs.

Lesson content

INSTRUMENT AND INDICATION SYSTEMS

Pressure gauge

- Different types, design, operation, characteristics, accuracy
- Temperature sensing
- Different types, design, operation, characteristics, accuracy

Fuel gauge

- Different types, design, operation, characteristics, accuracy
- Flow meter
- Different types, design, operation, characteristics, accuracy

Tachometer

- Different types, design, operation, characteristics, accuracy

Transponder

- Design, operation, characteristics, accuracy

MEASUREMENT OF AERODYNAMIC PARAMETERS

- Pressure measurement
- Static pressure, dynamic pressure, density: definitions
- Design, operation, errors, accuracy

Temperature measurement:

- Explain that the temperature shown is not always the true temperature and give example.

Altimeter

- Standard atmosphere
- The different barometric references (QNH, QFE and 1013.25)
- Height, indicated altitude, true altitude, pressure altitude and density altitude
- Design, operation, errors, accuracy

Vertical Speed Indicator

- Design, operation, errors, accuracy

Airspeed Indicator

- The different speeds IAS, CAS, TAS: definition, usage and relationships

- Design, operation, errors, accuracy

MAGNETISM – DIRECT READING COMPASS

- Earth magnetic field
- Direct reading compass
- Design, operation, data processing, accuracy, deviation
- Turning and acceleration errors

Gyroscope: basic principles

- Definitions, design, Fundamental properties, Drifts

Turn and bank indicator

- Design, operation, errors

Attitude indicator

- Design, operation, errors, accuracy

Directional gyroscope

- Design, operation, errors, accuracy

COMMUNICATION SYSTEMS

- Transmission modes: VHF, HF, Satcom
- Principles, bandwidth, operational limitations, use

Voice communication

- Definitions, general, applications

Warning Systems

- General Warning systems
- Explain the difference between yellow and red warning, and different system that use light, sound or text.
- Stall warning
- Design, operation, indications and alarms

INTEGRATED INSTRUMENTS – ELECTRONIC DISPLAYS

Electronic Flight Information system

- Explain the function of PFD and ND
- Explain how to fly with PFD and ND
- Explain how electric laser gyros are affected by loss of power.

TEM topics:

- Recognition of system failures, appropriate measures

Progress testing:

PT AGK I Lesson 01 performed at lessons end.

Interconnection:

- Recognition of/recovery from system failures.

Mandatory student preparation:

- Flygcert AGK I CBT chapters 1-5

AIRCRAFT GENERAL KNOWLEDGE – INSTRUMENTATION – Lesson 1

Learning Objectives

- 022 00 00 00 AIRCRAFT GENERAL KNOWLEDGE – INSTRUMENTATION
- 022 01 00 00 INSTRUMENT AND INDICATION SYSTEMS
 - 022 01 01 00 Pressure gauge
 - Different types, design, operation, characteristics, accuracy
 - 022 01 02 00 Temperature sensing
 - Different types, design, operation, characteristics, accuracy
 - 022 01 03 00 Fuel gauge
 - Different types, design, operation, characteristics, accuracy
 - 022 01 04 00 Flow meter
 - Different types, design, operation, characteristics, accuracy
 - 022 01 05 00 Tachometer
 - Different types, design, operation, characteristics, accuracy
 - 022 01 06 00 Transponder
 - Design, operation, characteristics, accuracy
- 022 02 00 00 MEASUREMENT OF AERODYNAMIC PARAMETERS
 - 022 02 01 00 Pressure measurement
 - Static pressure, dynamic pressure, density: definitions
 - Design, operation, errors, accuracy
 - 022 02 02 00 Temperature measurement:
 - Explain that the temperature shown is not always the true temperature and give example.
 - 022 02 04 00 Altimeter
 - Standard atmosphere
 - The different barometric references (QNH, QFE and 1013.25)
 - Height, indicated altitude, true altitude, pressure altitude and density altitude
 - Design, operation, errors, accuracy
 - 022 02 05 00 Vertical Speed Indicator
 - Design, operation, errors, accuracy
 - 022 02 06 00 Airspeed Indicator
 - The different speeds IAS, CAS, TAS: definition, usage and relationships
 - Design, operation, errors, accuracy
- 022 03 00 00 MAGNETISM – DIRECT READING COMPASS
 - 022 03 01 00 Earth magnetic field
 - 022 03 03 00 Direct reading compass
 - Design, operation, data processing, accuracy, deviation
 - Turning and acceleration errors
- 022 04 00 00 GYROSCOPIC INSTRUMENTS
 - 022 04 01 00 Gyroscope: basic principles
 - Definitions, design, Fundamental properties, Drifts
 - 022 04 02 00 Turn and bank indicator
 - Design, operation, errors
 - 022 04 03 00 Attitude indicator
 - Design, operation, errors, accuracy
 - 022 04 04 00 Directional gyroscope
 - Design, operation, errors, accuracy
- 022 10 00 00 COMMUNICATION SYSTEMS
 - 022 10 01 00 Transmission modes: VHF, HF, Satcom
 - Principles, bandwidth, operational limitations, use
 - 022 10 02 00 Voice communication
 - Definitions, general, applications
- 022 12 00 00 Warning Systems
 - 022 12 02 00 General Warning systems
 - Explain the difference between yellow and red warning, and different system that use light, sound or text.
 - 022 12 03 00 Stall warning
 - Design, operation, indications and alarms
- 022 13 00 00 INTEGRATED INSTRUMENTS – ELECTRONIC DISPLAYS

022 13 01 00 Electronic Flight Information system

- Explain the function of PFD and ND

- Explain how to fly with PFD and ND

Explain how electric laser gyros are affected by loss of power.

Mass and balance – Lesson 1

Overview

031 00 00 00 MASS AND BALANCE – AEROPLANES/HELICOPTERS
031 01 00 00 PURPOSE OF MASS AND BALANCE CONSIDERATIONS
031 02 00 00 LOADING
031 03 00 00 FUNDAMENTALS OF CG CALCULATIONS
031 04 00 00 MASS AND BALANCE DETAILS OF AIRCRAFT
031 05 00 00 DETERMINATION OF CG POSITION

Repetition
School Test
Preparation for Exam

Ref.:

1. SSFA LO PPL(A) 031 Mass and Balance
2. KSAB PPL Prestanda
3. KSAB L1P flyghandbok
4. Oxford or Jeppesen “Mass and Balance”, chapters included in PPL(A) LO requirements.
5. “Mass and Balance”, chapters included in PPL(A) LO requirements.
6. SSFA Lecture presentations “Mass and Balance”

Setting:
Classroom

Time allocated:
Instructor led study: 4 hrs.

Lesson content

Mass and balance considerations

- Mass limitations
- Structural limitations
- Performance limitations
- Centre of gravity (CG) limitations
- CG importance in regard to stability and controllability
- CG importance in regard to performance

LOADING

- Terminology
- Mass terms
- Load terms (including Fuel Terms)
- Mass limits
- Structural limitations
- Performance limitations
- Baggage compartment limitations
- Mass calculations
- Maximum masses for Takeoff and Landing
- Use of standard masses for passengers, baggage and crew

FUNDAMENTALS OF CG CALCULATIONS

- Definition of center of gravity
- Conditions of equilibrium (Balance of Forces and Balance of Moments)
- Basic calculations of CG

MASS AND BALANCE DETAILS OF AIRCRAFT

- Contents of mass and balance documentation
- Datum, moment arm
- CG position as distance from datum
- Extraction of basic mass and balance data from aircraft documentation
- Basic Empty Mass (BEM)
- CG position and/or moment at BEM

- Deviations from standard configuration

DETERMINATION OF CG POSITION

- Methods
- Mathematic method
- Graphic method
- Load and Trim Sheet
- General considerations
- Load sheet and CG envelope for light aeroplanes

TEM topics:

- Recognition of system failures, mechanical failures, electrical failures.

Progress testing:

PT M&B Lesson 01 performed at lessons end.

Interconnection:

- Recognition of/recovery from system failures.

Mandatory student preparation:

- Flygcert M&B CBT chapters 1-3

Mass and balance – Lesson 1

Learning Objectives

031 00 00 00 Mass and balance – Aeroplanes
031 01 00 00 Mass and balance considerations
031 01 01 00 Mass limitations
031 01 01 01 Structural limitations
031 01 01 02 Performance limitations
031 01 02 00 Centre of gravity (CG) limitations
031 01 02 01 CG importance in regard to stability and controllability
031 01 02 02 CG importance in regard to performance
031 02 00 00 LOADING
031 02 01 00 Terminology
031 02 01 01 Mass terms
031 02 01 02 Load terms (including Fuel Terms)
031 02 02 00 Mass limits
031 02 02 01 Structural limitations
031 02 02 02 Performance limitations
031 02 02 03 Baggage compartment limitations
031 02 03 00 Mass calculations
031 02 03 01 Maximum masses for Takeoff and Landing
031 02 03 03 Use of standard masses for passengers, baggage and crew
031 03 00 00 FUNDAMENTALS OF CG CALCULATIONS
031 03 01 00 Definition of center of gravity
031 03 02 00 Conditions of equilibrium (Balance of Forces and Balance of Moments)
031 03 03 00 Basic calculations of CG
031 04 00 00 MASS AND BALANCE DETAILS OF AIRCRAFT
031 04 01 00 Contents of mass and balance documentation
031 04 01 01 Datum, moment arm
031 04 01 02 CG position as distance from datum
031 04 03 00 Extraction of basic mass and balance data from aircraft documentation
031 04 03 01 Basic Empty Mass (BEM)
031 04 03 02 CG position and/or moment at BEM
031 04 03 03 Deviations from standard configuration
031 05 00 00 DETERMINATION OF CG POSITION
031 05 01 00 Methods
031 05 01 01 Mathematic method
031 05 01 02 Graphic method
031 05 02 00 Load and Trim Sheet
031 05 02 01 General considerations
031 05 02 02 Load sheet and CG envelope for light aeroplanes

Flight Performance – Lesson 1

Overview

032 00 00 00 PERFORMANCE – AEROPLANES

032 01 00 00 GENERAL

032 02 00 00 Single Engine Piston (SEP)

Ref.:

1. SSFA LO PPL(A) 032 Performance
2. KSAB PPL Prestanda
3. KSAB L1P flyghandbok
4. Oxford or Jeppesen “Performance”, chapters included in PPL(A) LO requirements.
5. “Performance”, chapters included in PPL(A) LO requirements.
6. SSFA Lecture presentations “Performance”

Setting:

Classroom

Time allocated:

Instructor led study: 4 hrs.

Lesson content

PERFORMANCE – AEROPLANES

- INTRODUCTION
- Performance
- Phases of flight
- factors affecting aircraft performance.
- Performance Classes
- Gradients (rising and decreasing)

TEM topics:

- Charts, interpreting and common errors

Progress testing:

PT PER Lesson 01 performed at lessons end.

Interconnection:

- Runway considerations, weather considerations

Mandatory student preparation:

- Flygcert PERFORMANCE CBT chapters 1-3

Flight Performance – Lesson 1

Learning Objectives

032 00 00 00 PERFORMANCE – AEROPLANES
032 01 00 00 INTRODUCTION
032 01 02 00 Performance
032 01 02 01 Phases of flight
032 01 02 03 factors affecting aircraft performance.
032 01 02 04 Performance Classes
032 01 02 05 Gradients (rising and decreasing)

Flight Performance – Lesson 2

Overview

032 02 00 00 SINGLE ENGINE AEROPLANES

Repetition
School Test
Preparation for Exam

Ref.:

1. SSFA LO PPL(A) 032 Performance
2. KSAB PPL Prestanda
3. KSAB L1P flyghandbok
4. Oxford or Jeppesen “Performance”, chapters included in PPL(A) LO requirements.
5. “Performance”, chapters included in PPL(A) LO requirements.
6. SSFA Lecture presentations “Performance”

Setting:

Classroom

Time allocated:

Instructor led study: 4 hrs.

Lesson content

SINGLE ENGINE AEROPLANES

- Definitions of terms and speeds
- Pitch angle and level flight attitude
- Use of aeroplane performance data
- Take-off performance
- Climb performance
- Cruise Performance
- Landing Performance

TEM topics:

Charts, interpreting and common errors

Progress testing:

PT PER Lesson 02 performed at lessons end.

Interconnection:

Runway/weather/aircraft performance considerations

Mandatory student preparation:

Flygcert PERFORMANCE CBT chapters 3-6

Flight Performance – Lesson 2

Learning Objectives

032 02 00 00 SINGLE ENGINE AEROPLANES
032 02 01 00 Definitions of terms and speeds
032 02 04 00 Pitch angle and level flight attitude
032 02 05 00 Use of aeroplane performance data
032 02 05 01 Take-off performance
032 02 05 02 Climb performance
032 02 05 03 Cruise Performance
032 02 05 05 Landing Performance

Flight Planning and Flight Monitoring – Lesson 1

Overview

033 00 00 00 FLIGHT PLANNING AND FLIGHT MONITORING

033 01 00 00 FLIGHT PLANNING FOR VFR FLIGHTS

033 01 02 00 Documentation on board

033 03 00 00 FUEL PLANNING

Ref.:

1. SSFA LO PPL(A) 033 Flight Planning
2. KSAB PPL Genomförande och Planering av flygningar
3. KSAB L1P flyghandbok
4. Oxford or Jeppesen “Flight Planning”, chapters included in PPL(A) LO requirements.
5. “Flight Planning”, chapters included in PPL(A) LO requirements.
6. SSFA Lecture presentations “Flight Planning”

Setting:

Classroom

Time allocated:

Instructor led study: 4 hrs.

Lesson content

FLIGHT PLANNING FOR VFR FLIGHTS

- VFR Navigation plan
- Routes, airfields, heights and altitudes from VFR charts
- Courses and distances from VFR charts
- Aerodrome Charts and Aerodrome Directory
- Information for communication and radio aids
- Completion of operational flight plan
- Documents to be onboard.

FUEL PLANNING

- General knowledge
- Calculation of required fuel for flight
- Calculation of extra fuel
- Completion of the fuel section of the navigation plan (fuel log) and calculation of total fuel

TEM topics:

- Charts, interpreting and common errors

Progress testing:

PT FP&M Lesson 01 performed at lessons end.

Interconnection:

- Dep/Dest/Route altitudes

Mandatory student preparation:

- Flygcert FP&M CBT chapters 1-3

Flight Planning and Flight Monitoring – Lesson 1

Learning Objectives

033 00 00 00 FLIGHT PLANNING AND FLIGHT MONITORING
033 01 00 00 FLIGHT PLANNING FOR VFR FLIGHTS
033 01 01 00 VFR Navigation plan
033 01 01 01 Routes, airfields, heights and altitudes from VFR charts
033 01 01 02 Courses and distances from VFR charts
033 01 01 03 Aerodrome Charts and Aerodrome Directory
033 01 01 04 Information for communication and radio aids
033 01 01 05 Completion of operational flight plan
033 01 02 00 Documents to be onboard.
033 03 00 00 FUEL PLANNING
033 03 01 00 General knowledge
033 03 01 01 Calculation of required fuel for flight
033 03 01 02 Calculation of extra fuel
033 03 01 03 Completion of the fuel section of the navigation plan (fuel log) and calculation of total fuel

Flight Planning and Flight Monitoring – Lesson 2

Overview

033 04 00 00 PRE-FLIGHT PREPARATION

033 05 00 00 ICAO FLIGHT PLAN (ATS Flight Plan)

033 06 00 00 FLIGHT MONITORING AND IN FLIGHT RE PLANNING

Repetition

School Test

Preparation for Exam

Ref.:

1. SSFA LO PPL(A) 033 Flight Planning
2. KSAB PPL Genomförande och Planering av flygningar
3. KSAB L1P flyghandbok
4. Oxford or Jeppesen “Flight Planning”, chapters included in PPL(A) LO requirements.
5. “Flight Planning”, chapters included in PPL(A) LO requirements.
6. SSFA Lecture presentations “Flight Planning”

Setting:

Classroom

Time allocated:

Instructor led study: 4 hrs.

Lesson content

PRE-FLIGHT PREPARATION

- AIP and NOTAM briefing
- Ground facilities and services
- Departure, destination and alternate aerodromes
- Airway routings and airspace structure
- Meteorological information
- Extraction and analysis of relevant data from meteorological documents

ICAO FLIGHT PLAN (ATS Flight Plan)

- ATS flight plan
- Format of ATS flight plan
- Summary of ATS flight plan
- Submission of the flight plan

FLIGHT MONITORING AND IN-FLIGHT RE-PLANNING

- Flight monitoring
- Monitoring of track and time
- In flight fuel management
- In flight re-planning in case of deviation from planned data

TEM topics:

- Metrological considerations

Progress testing:

PT FP&M Lesson 02 performed at lessons end.

Interconnection:

- Routing, weather considerations, departure/alternate airdromes

Mandatory student preparation:

- Flygcert FP&M CBT chapters 2-4

Flight Planning and Flight Monitoring – Lesson 2

Learning Objectives

033 04 00 00 PRE-FLIGHT PREPARATION
033 04 01 00 AIP and NOTAM briefing
033 04 01 01 Ground facilities and services
033 04 01 02 Departure, destination and alternate aerodromes
033 04 01 03 Airway routings and airspace structure
033 04 02 00 Meteorological information
033 04 02 01 Extraction and analysis of relevant data from meteorological documents
033 05 00 00 ICAO FLIGHT PLAN (ATS Flight Plan)
033 05 01 00 ATS flight plan
033 05 01 01 Format of ATS flight plan
033 05 01 02 Summary of ATS flight plan
033 05 03 03 Submission of the flight plan
033 06 00 00 FLIGHT MONITORING AND IN-FLIGHT RE-PLANNING
033 06 01 00 Flight monitoring
033 06 01 01 Monitoring of track and time
033 06 01 02 In flight fuel management
033 06 02 00 In flight re-planning in case of deviation from planned data

HUMAN PERFORMANCE – Lesson 1

Overview

040 00 00 00 HUMAN PERFORMANCE

040 01 00 00 HUMAN FACTORS: BASIC CONCEPTS

040 02 02 00 Man and Environment

Ref.:

1. SSFA LO PPL(A) 040 Human Performance and Limitations
2. KSAB PPL Människans begränsningar och förutsättningar
3. Oxford or Jeppesen “Human Performance and Limitations”, chapters included in PPL(A) LO requirements.
4. SSFA Lecture presentations 040 Human Performance and Limitations

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

HUMAN FACTORS: BASIC CONCEPTS

- Human Factors in aviation

BASIC AVIATION PHYSIOLOGY AND HEALTH MAINTENANCE

The atmosphere

- composition
- gas Laws

Respiratory and circulatory systems

- oxygen requirement of tissues
- functional anatomy
- main forms of hypoxia (hypoxic and anemic)
- sources, effects and counter measures of carbon monoxide
- counter measures, hypoxia
- symptoms of hypoxia
- hyperventilation
- the effects of accelerations on the circulatory system
- hypertension and coronary heart disease

Vision

- functional anatomy
- visual field, foveal and peripheral vision
- binocular and monocular vision
- monocular vision cues
- night vision
- visual scanning and detection techniques and importance of “lookout”
- defective vision

Hearing and Equilibrium

- descriptive and functional anatomy
- flight related hazards to hearing
- hearing loss
- functional anatomy
- motion and acceleration
- motion sickness
- spatial disorientation: forms, recognition, avoidance
- illusions: forms, recognition, avoidance
- physical origin
- physiological origin
- psychological origin
- approach and landing problems

Personal hygiene

- personal fitness

Body rhythm and sleep

- rhythm disturbances
- symptoms, effects, management

Problem areas for pilots

- common minor ailments including cold, influenza and gastro intestinal upset
- entrapped gases, barotrauma, (scuba diving)
- obesity
- food hygiene
- infectious diseases
- nutrition
- various toxic gases and materials

Intoxication

- prescribed medication
- tobacco
- alcohol and drugs
- caffeine
- self-medication

TEM topics:

- Fatigue, illness, alcohol, stress

Progress testing:

PT HPL Lesson 01 performed at lessons end.

Interconnection:

- Multitasking, stress, physiological effects of flying

Mandatory student preparation:

- Flygcert HPL CBT chapters 1-5

HUMAN PERFORMANCE – Lesson 1

Learning Objectives

- 040 00 00 00 HUMAN PERFORMANCE
- 040 01 00 00 HUMAN FACTORS: BASIC CONCEPTS
- 040 01 01 00 Human Factors in aviation
- 040 02 00 00 BASIC AVIATION PHYSIOLOGY AND HEALTH MAINTENANCE
- 040 02 01 01 The atmosphere
 - composition
 - gas Laws
- 040 02 01 02 Respiratory and circulatory systems
 - oxygen requirement of tissues
 - functional anatomy
 - main forms of hypoxia (hypoxic and anemic)
 - sources, effects and counter measures of carbon monoxide
 - counter measures, hypoxia
 - symptoms of hypoxia
 - hyperventilation
 - the effects of accelerations on the circulatory system
 - hypertension and coronary heart disease
- 040 02 02 00 Man and Environment
- 040 02 02 01 Central, peripheral and autonomic nervous systems
- 040 02 02 02 Vision
 - functional anatomy
 - visual field, foveal and peripheral vision
 - binocular and monocular vision
 - monocular vision cues
 - night vision
 - visual scanning and detection techniques and importance of “lookout”
 - defective vision
- 040 02 02 03 Hearing and Equilibrium
 - descriptive and functional anatomy
 - flight related hazards to hearing
 - hearing loss
 - functional anatomy
 - motion and acceleration
 - motion sickness
 - spatial disorientation: forms, recognition, avoidance
 - illusions: forms, recognition, avoidance
 - physical origin
 - physiological origin
 - psychological origin
 - approach and landing problems
- 040 02 03 00 Health and hygiene
- 040 02 03 01 Personal hygiene
 - personal fitness
 - Body rhythm and sleep
 - rhythm disturbances
 - symptoms, effects, management
- Problem areas for pilots
 - common minor ailments including cold, influenza and gastro intestinal upset
 - entrapped gases, barotrauma, (scuba diving)
 - obesity
 - food hygiene
 - infectious diseases
 - nutrition
 - various toxic gases and materials
- Intoxication
 - prescribed medication

- tobacco
- alcohol and drugs
- caffeine
- self-medication

HUMAN PERFORMANCE – Lesson 2

Overview

040 03 00 00 BASIC AVIATION PSYCHOLOGY

Repetition

School Test

Preparation for Exam

Ref.:

1. SSFA LO PPL(A) 040 Human Performance and Limitations
2. KSAB PPL Människans begränsningar och förutsättningar
3. Oxford or Jeppesen “Human Performance and Limitations”, chapters included in PPL(A) LO requirements.
4. SSFA Lecture presentations 040 Human Performance and Limitations

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

Human information processing

- Attention and vigilance
- selectivity of attention
- divided attention

Perception

- perceptual illusions
- subjectivity of perception
- processes of perception

Memory

- sensory memory
- working/short term memory
- long term memory to include motor memory (skills)

Human error and reliability

- Reliability of human behavior
- Error generation
- social environment (group, organization)

Decision making

- Decision making concepts.
- identification of the target
- collection of information
- risk assessment
- development of alternative measures
- evaluation of alternative measures
- decision-making
- implementation
- impact analysis
- evaluation.

Avoiding and managing errors: cockpit management.

- Safety awareness
- risk area awareness
- situational awareness

Communication: verbal and nonverbal communication

- Human behavior
- Personality and attitudes
- development
- environmental influences

Identification of hazardous attitudes (error proneness)

Human overload and underload

Arousal**Stress and fatigue**

- definition(s)
- anxiety and stress
- effects of stress

Fatigue and stress management

- types, causes and symptoms of fatigue
- effects of fatigue
- coping strategies
- management techniques
- health and fitness programmed

TEM topics:

- Fatigue, illness, alcohol, stress

Progress testing:

PT HPL Lesson 02 performed at lessons end.

Interconnection:

- Multitasking, stress, physiological effects of flying

Mandatory student preparation:

- Flygcert HPL CBT chapters 5-10

HUMAN PERFORMANCE – Lesson 2

Learning Objectives

- 040 03 00 00 BASIC AVIATION PSYCHOLOGY
- 040 03 01 00 Human information processing
- 040 03 01 01 Attention and vigilance
 - selectivity of attention
 - divided attention
- 040 03 01 02 Perception
 - perceptual illusions
 - subjectivity of perception
 - processes of perception
- 040 03 01 03 Memory
 - sensory memory
 - working/short term memory
 - long term memory to include motor memory (skills)
- 040 03 02 00 Human error and reliability
 - Reliability of human behavior
 - Error generation
 - social environment (group, organization)
- 040 03 03 00 Decision making
 - Decision making concepts.
 - identification of the target
 - collection of information
 - risk assessment
 - development of alternative measures
 - evaluation of alternative measures
 - decision-making
 - implementation
 - impact analysis
 - evaluation.
- 040 03 04 00 Avoiding and managing errors: cockpit management.
 - Safety awareness
 - risk area awareness
 - situational awareness
- 040 03 04 04 Communication: verbal and nonverbal communication
- 040 03 05 00 Human behavior
- 040 03 05 01 Personality and attitudes
 - development
 - environmental influences
- 040 03 05 03 Identification of hazardous attitudes (error proneness)
- 040 03 06 00 Human overload and underload
- 040 03 06 01 Arousal
- 040 03 06 02 Stress and fatigue
 - definition(s)
 - anxiety and stress
 - effects of stress
- Fatigue and stress management
 - types, causes and symptoms of fatigue
 - effects of fatigue
 - coping strategies
 - management techniques
 - health and fitness programmed

METEOROLOGY – Lesson 1

Overview

Exams in metrology require basic knowledge of geography. This includes names and positions of larger cities, oceans and lakes in Sweden as well as a broad overview of the rest of the Nordic countries.

050 01 00 00 THE ATMOSPHERE
050 01 02 00 AIR TEMPERATURE
050 01 03 00 ATMOSPHERIC PRESSURE
050 01 04 00 AIR DENSITY
050 01 05 01 ICAO STANDARD ATMOSPHERE
050 01 06 00 ALTIMETRY

Ref.:

1. SSFA LO PPL(A) 050 Meteorology
2. KSAB PPL Meteorology
3. Oxford or Jeppesen “Meteorology”, chapters included in PPL(A) LO requirements.
4. SSFA Lecture presentations 050 Meteorology

Setting:

Classroom

Time allocated:

Instructor led study: 4 hrs.

Lesson content

THE ATMOSPHERE

- Composition, extent, vertical division
- Structure of the atmosphere
- Troposphere
- Air temperature
- Definition and units
- Vertical distribution of temperature
- Transfer of heat
- Lapse rates, stability and instability
- Development of inversions, types of inversions
- Temperature near the earth 's surface, surface effects, diurnal and seasonal variation, effect of clouds, effect of wind
- Atmospheric pressure
- Barometric pressure, isobars
- Pressure variation with height
- Reduction of pressure to mean sea level
- Relationship between surface pressure centers and pressure centers aloft
- Air density
- Relationship between pressure, temperature and density
- ICAO Standard Atmosphere (ISA)
- Altimetry
- Terminology and definitions
- Altimeter and altimeter settings
- Calculations
- Effect of accelerated airflow due to topography

TEM topics:

- Effects of high altitude flying

Progress testing:

PT MET Lesson 01 performed at lessons end.

Interconnection:

- Altimetry, selecting appropriate altitude

Mandatory student preparation:

- Flygcert CBT MET chapters 1-5

METEOROLOGY – Lesson 1

050 00 00 00 METEOROLOGY
050 01 00 00 THE ATMOSPHERE
050 01 01 00 Composition, extent, vertical division
050 01 01 01 Structure of the atmosphere
050 01 01 02 Troposphere
050 01 02 00 Air temperature
050 01 02 01 Definition and units
050 01 02 02 Vertical distribution of temperature
050 01 02 03 Transfer of heat
050 01 02 04 Lapse rates, stability and instability
050 01 02 05 Development of inversions, types of inversions
050 01 02 06 Temperature near the earth 's surface, surface effects, diurnal and seasonal variation, effect of clouds, effect of wind
050 01 03 00 Atmospheric pressure
050 01 03 01 Barometric pressure, isobars
050 01 03 02 Pressure variation with height
050 01 03 03 Reduction of pressure to mean sea level
050 01 03 04 Relationship between surface pressure centers and pressure centers aloft
050 01 04 00 Air density
050 01 04 01 Relationship between pressure, temperature and density
050 01 05 00 ICAO Standard Atmosphere (ISA)
050 01 06 00 Altimetry
050 01 06 01 Terminology and definitions
050 01 06 02 Altimeter and altimeter settings
050 01 06 03 Calculations
050 01 06 04 Effect of accelerated airflow due to topography

METEOROLOGY – Lesson 2

Overview

050 02 00 00 WIND
050 03 00 00 THERMODYNAMICS
050 04 00 00 CLOUDS AND FOG
050 05 00 00 PRECIPITATION

Ref.:

1. SSFA LO PPL(A) 050 Meteorology
2. KSAB PPL Meteorology
3. Oxford or Jeppesen “Meteorology”, chapters included in PPL(A) LO requirements.
4. SSFA Lecture presentations 050 Meteorology

Setting:

Classroom

Time allocated:

Instructor led study: 4 hrs.

Lesson content

WIND

- Definition and measurement of wind
- Definition and measurement
- Primary cause of wind
- Primary cause of wind, pressure gradient, Coriolis force, gradient wind
- Variation of wind in the friction layer
- Effects of convergence and divergence
- General global circulation
General circulation around the world
- Local winds
Anabatic and katabatic winds, mountain and valley winds, Venturi effects, land and sea breezes
- Mountain waves (standing waves, lee waves)
- Origin and characteristics
- Turbulence
- Description and type of turbulence

THERMODYNAMICS

- Humidity
- Water vapor in the atmosphere
- Temperature/dew point, relative humidity
- Mixing ratio
- Change of state of aggregation
- Condensation, evaporation, sublimation, freezing and melting, latent heat
- Adiabatic processes
- Adiabatic processes, stability of the atmosphere

CLOUDS AND FOG

- Cloud formation and description
- Cooling by adiabatic expansion and by advection
- Cloud types and cloud classification
- Influence of inversions on cloud development
- Fog, mist, haze
- General aspects
- Radiation fog
- Advection fog
- Steaming fog
- Frontal fog
- Orographic fog (hill fog)

PRECIPITATION

- Development of precipitation
- Process of development of precipitation
- Types of precipitation
- Types of precipitation, relationship with cloud types

TEM topics:

- Wind considerations/potential dangers derived from

Progress testing:

PT MET Lesson 02 performed at lessons end.

Interconnection:

- Nose wind, cross wind, turbulence, wake

Mandatory student preparation:

- Flygcert CBT MET chapters 5-10

METEOROLOGY – Lesson 2

Learning Objectives

050 02 00 00 WIND
050 02 01 00 Definition and measurement of wind
050 02 01 01 Definition and measurement
050 02 02 00 Primary cause of wind
050 02 02 01 Primary cause of wind, pressure gradient, Coriolis force, gradient wind
050 02 02 02 Variation of wind in the friction layer
050 02 02 03 Effects of convergence and divergence
050 02 03 00 General global circulation
050 02 03 01 General circulation around the world
050 02 04 00 Local winds
050 02 04 01 Anabatic and katabatic winds, mountain and valley winds, Venturi effects, land and sea breezes
050 02 05 00 Mountain waves (standing waves, lee waves)
050 02 05 01 Origin and characteristics
050 02 06 00 Turbulence
050 02 06 01 Description and type of turbulence
050 03 00 00 THERMODYNAMICS
050 03 01 00 Humidity
050 03 01 01 Water vapor in the atmosphere
050 03 01 03 Temperature/dew point, relative humidity
050 03 01 04 Mixing ratio
050 03 02 00 Change of state of aggregation
050 03 02 01 Condensation, evaporation, sublimation, freezing and melting, latent heat
050 03 03 00 Adiabatic processes
050 03 03 01 Adiabatic processes, stability of the atmosphere
050 04 00 00 CLOUDS AND FOG
050 04 01 00 Cloud formation and description
050 04 01 01 Cooling by adiabatic expansion and by advection
050 04 01 02 Cloud types and cloud classification
050 04 01 03 Influence of inversions on cloud development
050 04 02 00 Fog, mist, haze
050 04 02 01 General aspects
050 04 02 02 Radiation fog
050 04 02 03 Advection fog
050 04 02 04 Steaming fog
050 04 02 05 Frontal fog
050 04 02 06 Orographic fog (hill fog)
050 05 00 00 PRECIPITATION
050 05 01 00 Development of precipitation
050 05 01 01 Process of development of precipitation
050 05 02 00 Types of precipitation
050 05 02 01 Types of precipitation, relationship with cloud types

METEOROLOGY – Lesson 3

Overview

050 06 00 00 AIR MASSES AND FRONTS

050 07 00 00 PRESSURE SYSTEMS

050 08 00 00 CLIMATOLOGY

Ref

1. SSFA LO PPL(A) 050 Meteorology
2. KSAB PPL Meteorology
3. Oxford or Jeppesen “Meteorology”, chapters included in PPL(A) LO requirements.
4. SSFA Lecture presentations 050 Meteorology

Setting:

Classroom

Time allocated:

Instructor led study: 4 hrs.

Lesson content

AIR MASSES AND FRONTS

- Description, classification and source regions of air masses.
- Changes of air masses properties
- Fronts
- General aspects
- Warm front, associated clouds, and weather
- Cold front, associated clouds, and weather
- Warm sector, associated clouds, and weather
- Weather behind the cold front
- Occlusions, associated clouds, and weather
- Stationary front, associated clouds, and weather
- Movement of fronts and pressure systems, life cycle
- Changes of meteorological elements at a frontal wave

PRESSURE SYSTEMS

- Anticyclone
- Anticyclones, types, general properties, cold and warm anticyclones, ridges and wedges, subsidence
- Non frontal depressions
- Thermal, orographic and polar depressions, troughs

CLIMATOLOGY

- Typical weather situations at our latitudes
- West location
- High pressure areas
- Low Air pressure gradient
- Climate zones
- General seasonal circulation in the troposphere

TEM topics:

- Frontal weather, high vs low pressure, high vs low temp

Progress testing:

PT MET Lesson 03 performed at lessons end.

Interconnection:

- Typical weather conditions,

Mandatory student preparation:

- Flygcert CBT MET chapters 10-14

METEOROLOGY – Lesson 3

Learning Objectives

050 06 00 00 AIR MASSES AND FRONTS

050 06 01 00 Air masses

050 06 01 01 Description, classification and source regions of air masses.

050 06 01 02 Changes of air masses properties

050 06 02 00 Fronts

050 06 02 01 General aspects

050 06 02 02 Warm front, associated clouds, and weather

050 06 02 03 Cold front, associated clouds, and weather

050 06 02 04 Warm sector, associated clouds, and weather

050 06 02 05 Weather behind the cold front

050 06 02 06 Occlusions, associated clouds, and weather

050 06 02 07 Stationary front, associated clouds, and weather

050 06 02 08 Movement of fronts and pressure systems, life cycle

050 06 02 09 Changes of meteorological elements at a frontal wave

050 07 00 00 PRESSURE SYSTEMS

050 07 02 00 Anticyclone

050 07 02 01 Anticyclones, types, general properties, cold and warm anticyclones, ridges and wedges, subsidence

050 07 03 00 Non frontal depressions

050 07 03 01 Thermal, orographic and polar depressions, troughs

050 08 00 00 CLIMATOLOGY

050 08 03 00 Typical weather situations at our latitudes

050 08 03 01 West location

050 08 03 02 High pressure areas

050 08 03 03 Low Air pressure gradient

050 08 04 00 Climate zones

050 08 04 01 General seasonal circulation in the troposphere

METEOROLOGY – Lesson 4

Overview

050 09 00 00 FLIGHT HAZARDS

050 10 00 00 METEOROLOGICAL INFORMATION

Ref.:

1. SSFA LO PPL(A) 050 Meteorology
2. KSAB PPL Meteorology
3. Oxford or Jeppesen “Meteorology”, chapters included in PPL(A) LO requirements.
4. SSFA Lecture presentations 050 Meteorology

Setting:

Classroom

Time allocated:

Instructor led study: 4 hrs.

Lesson content

FLIGHT HAZARDS

Icing

- Conditions for ice accretion
- Types of ice accretion
- Hazards of ice accretion, avoidance

Turbulence

- Effects on flight, avoidance
- Wind shear
- Definition of wind shear
- Weather conditions for wind shear
- Effects on flight, avoidance
- Thunderstorms
- Conditions for, and process of, development, forecast, location, type specification
- Structure of thunderstorms, life cycle, squall lines, electricity in the atmosphere, static charges
- Electrical discharges
- Development and effects of downbursts
- Thunderstorm avoidance

Inversions

- Influence on aircraft performance
- Hazards in mountainous areas
- Influence of terrain on clouds and precipitation, frontal passage
- Vertical movements, mountain wave, wind shear, turbulence, ice accretion
- Development and effect of valley inversions
- Visibility reducing phenomena.
- Reduction of visibility caused by precipitation and obscuration.
- Reduction of visibility caused by other phenomena.

METEOROLOGICAL INFORMATION

- Observation
- Surface observations
- Radiosonde observations
- Satellite observations
- Weather radar observations
- Aircraft observations and reporting

Weather charts

- Significant weather charts
- Surface charts

Information for flight planning

- Aviation weather messages
- Meteorological broadcasts for aviation
- Use of meteorological documents
- Meteorological warnings
- Meteorological services
- World area forecast system (WAFS) and meteorological offices.

TEM topics:

- Turbulence, wind shear, thunder storms

Progress testing:

PT MET Lesson 04 performed at lessons end.

Interconnection:

- Meteorological charts/information – fly no fly decisions

Mandatory student preparation:

- Flygcert CBT MET chapters 14-19

METEOROLOGY – Lesson 4

Learning Objectives

050 09 00 00 FLIGHT HAZARDS
050 09 01 00 Icing
050 09 01 01 Conditions for ice accretion
050 09 01 02 Types of ice accretion
050 09 01 03 Hazards of ice accretion, avoidance
050 09 02 00 Turbulence
050 09 02 01 Effects on flight, avoidance
050 09 03 00 Wind shear
050 09 03 01 Definition of wind shear
050 09 03 02 Weather conditions for wind shear
050 09 03 03 Effects on flight, avoidance
050 09 04 00 Thunderstorms
050 09 04 01 Conditions for, and process of, development, forecast, location, type specification
050 09 04 02 Structure of thunderstorms, life cycle, squall lines, electricity in the atmosphere, static charges
050 09 04 03 Electrical discharges
050 09 04 04 Development and effects of downbursts
050 09 04 05 Thunderstorm avoidance
050 09 06 00 Inversions
050 09 06 01 Influence on aircraft performance
050 09 08 00 Hazards in mountainous areas
050 09 08 01 Influence of terrain on clouds and precipitation, frontal passage
050 09 08 02 Vertical movements, mountain wave, wind shear, turbulence, ice accretion
050 09 08 03 Development and effect of valley inversions
050 09 09 00 Visibility reducing phenomena.
050 09 09 01 Reduction of visibility caused by precipitation and obscuration.
050 09 09 02 Reduction of visibility caused by other phenomena.
050 10 00 00 METEOROLOGICAL INFORMATION
050 10 01 00 Observation
050 10 01 01 Surface observations
050 10 01 02 Radiosonde observations
050 10 01 03 Satellite observations
050 10 01 04 Weather radar observations
050 10 01 05 Aircraft observations and reporting
050 10 02 00 Weather charts
050 10 02 01 Significant weather charts
050 10 02 02 Surface charts
050 10 03 00 Information for flight planning
050 10 03 01 Aviation weather messages
050 10 03 02 Meteorological broadcasts for aviation
050 10 03 03 Use of meteorological documents
050 10 03 04 Meteorological warnings
050 10 04 00 Meteorological services
050 10 04 01 World area forecast system (WAFS) and meteorological offices.
Repetition
School Test
Preparation for Exam

General Navigation – Lesson 1

Overview

060 00 00 00 NAVIGATION

061 00 00 00 GENERAL NAVIGATION

061 01 00 00 BASICS OF NAVIGATION

061 02 00 00 MAGNETISM AND COMPASSES

Ref.:

1. SSFA LO PPL(A) 060 General Navigation
2. KSAB PPL Navigation
3. Oxford or Jeppesen “General Navigation”, chapters included in PPL(A) LO requirements.
4. SSFA Lecture presentations 061 General Navigation

Setting:

Classroom

Time allocated:

Instructor led study: 4 hrs.

Lesson content

BASICS OF NAVIGATION

- The solar system
- Earth movement in the solar system
- The earth
- Great circle, small circle, rhumb line
- Latitude, difference of latitude
- Longitude, difference of longitude
- Use of latitude and longitude coordinates system

Time

- solar time
- UTC
- LMT
- Standard times
- Dateline
- Definition of sunrise, sunset and civil twilight

Directions

- True north (True Track, True Bearing, True Heading)
- Magnetic North (Magnetic Track, Magnetic Bearing, Magnetic Heading)
- Deviation (Compass North)
- Earth Magnetic Field

Distance

- Units of distance and height used in navigation.
- Conversion of units
- Relationship between nautical miles and minutes of latitude and minutes of longitude

MAGNETISM AND COMPASSES

- General Principles of compass
- Terrestrial magnetism, Variation annual change-
- Resolution of the earth's total magnetic force into vertical and horizontal components
- Aircraft magnetism
- The resulting magnetic fields
- Keeping magnetic materials clear of the compass

TEM topics:

- Lost procedures, equipment failure procedures

Progress testing:

PT GNAV Lesson 01 performed at lessons end.

Interconnection:

- Principles of compass

Mandatory student preparation:

- Flygcert CBT GNAV chapters 1-5

General Navigation – Lesson 1

Learning Objectives

060 00 00 00 NAVIGATION
061 00 00 00 GENERAL NAVIGATION
061 01 00 00 BASICS OF NAVIGATION
061 01 01 00 The solar system
061 01 01 01 Earth movement in the solar system
061 01 02 00 The earth
061 01 02 01 Great circle, small circle, rhumb line
061 01 02 03 Latitude, difference of latitude
061 01 02 04 Longitude, difference of longitude
061 01 02 05 Use of latitude and longitude coordinates system
061 01 03 00 Time
061 01 03 01 solar time
061 01 03 02 UTC
061 01 03 03 LMT
061 01 03 04 Standard times
061 01 03 05 Dateline
061 01 03 06 Definition of sunrise, sunset and civil twilight
061 01 04 00 Directions
061 01 04 01 True north (True Track, True Bearing, True Heading)
061 01 04 02 Magnetic North (Magnetic Track, Magnetic Bearing, Magnetic Heading)
061 01 04 03 Deviation (Compass North)
061 01 04 04 Earth Magnetic Field
061 01 05 00 Distance
061 01 05 01 Units of distance and height used in navigation.
061 01 05 02 Conversion of units
061 01 05 03 Relationship between nautical miles and minutes of latitude and minutes of longitude
061 02 00 00 MAGNETISM AND COMPASSES
061 02 01 00 General Principles of compass
061 02 01 01 Terrestrial magnetism, Variation annual change-
061 02 01 02 Resolution of the earth's total magnetic force into vertical and horizontal components
061 02 02 00 Aircraft magnetism
-The resulting magnetic fields
-Keeping magnetic materials clear of the compass

General Navigation – Lesson 2

Overview

061 03 00 00 CHARTS
061 04 00 00 DEAD RECKONING NAVIGATION (DR)
061 05 00 00 IN FLIGHT NAVIGATION

Repetition
School Test
Preparation for Exam

Ref.:

1. SSFA LO PPL(A) 060 General Navigation
2. KSAB PPL Navigation
3. Oxford or Jeppesen “General Navigation”, chapters included in PPL(A) LO requirements.
4. SSFA Lecture presentations 061 General Navigation

Setting:
Classroom

Time allocated:
Instructor led study: 4 hrs.

Lesson content

CHARTS

- General properties of different types of projections
- Direct Mercator
- Lambert conformal conic
- The representation of meridians, parallels, great circles and rhumb lines
- Direct Mercator
- Lambert conformal conic
- The use of aeronautical charts
- Plotting positions
- Methods of indicating scale and relief (ICAO topographical chart)
- Chart symbols
- Measuring tracks and distances
- Plotting bearings and distances

DEAD RECKONING NAVIGATION (DR)

- Basis of dead reckoning
- Track
- Heading (compass, magnetic, true)
- Wind velocity
- Airspeed (IAS, CAS, TAS)
- Groundspeed
- ETA
- Drift, wind correction angle
- DR position fix
- Use of the navigational computer
- Speed
- Time
- Distance
- Fuel consumption
- Conversions
- Airspeed
- Wind velocity
- True altitude
- The triangle of velocities
- Heading

- Ground speed
- Wind velocity
- Track and drift angle
- Wind Correction angle
- Measurement of DR elements
- Height, time and distance calculation

IN FLIGHT NAVIGATION

- Use of visual observations and application to in flight navigation
- Navigation in cruising flight, use of fixes to revise navigation data
- Ground speed revision
- Off track corrections
- Calculation of wind speed and direction
- ETA revisions
- Flight Log

TEM topics:

- Fuel consumption, alternates

Progress testing:

PT GNAV Lesson 02 performed at lessons end.

Interconnection:

- Charts, interpretation and follow up

Mandatory student preparation:

- Flygcert CBT GNAV chapters 6-10

General Navigation – Lesson 2

Learning Objectives

061 03 00 00 CHARTS
061 03 01 00 General properties of different types of projections
061 03 01 01 Direct Mercator
061 03 01 02 Lambert conformal conic
061 03 02 00 The representation of meridians, parallels, great circles and rhumb lines
061 03 02 01 Direct Mercator
061 03 02 02 Lambert conformal conic
061 03 03 00 The use of aeronautical charts
061 03 03 01 Plotting positions
061 03 03 02 Methods of indicating scale and relief (ICAO topographical chart)
061 03 03 03 Chart symbols
061 03 03 04 Measuring tracks and distances
061 03 03 05 Plotting bearings and distances
061 04 00 00 DEAD RECKONING NAVIGATION (DR)
061 04 01 00 Basis of dead reckoning
061 04 01 01 Track
061 04 01 02 Heading (compass, magnetic, true)
061 04 01 03 Wind velocity
061 04 01 04 Airspeed (IAS, CAS, TAS)
061 04 01 05 Groundspeed
061 04 01 06 ETA
061 04 01 07 Drift, wind correction angle
061 04 01 08 DR position fix
061 04 02 00 Use of the navigational computer
061 04 02 01 Speed
061 04 02 02 Time
061 04 02 03 Distance
061 04 02 04 Fuel consumption
061 04 02 05 Conversions
061 04 02 06 Airspeed
061 04 02 07 Wind velocity
061 04 02 08 True altitude
061 04 03 00 The triangle of velocities
-Heading
-Ground speed
-Wind velocity
-Track and drift angle
-Wind Correction angle
061 04 05 00 Measurement of DR elements
061 04 05 01 Height, time and distance calculation
061 05 00 00 IN FLIGHT NAVIGATION
061 05 01 00 Use of visual observations and application to in flight navigation
061 05 03 00 Navigation in cruising flight, use of fixes to revise navigation data
061 05 03 01 Ground speed revision
061 05 03 02 Off track corrections
061 05 03 03 Calculation of wind speed and direction
061 05 03 04 ETA revisions
061 05 04 00 Flight Log

NAVIGATION – RADIO NAVIGATION – Lesson 1

Overview

062 00 00 00 RADIO NAVIGATION
062 01 00 00 BASIC RADIO PROPAGATION THEORY
062 02 00 00 RADIO AIDS

Ref.:

1. SSFA LO PPL(A) 062 Radio Navigation
2. Oxford or Jeppesen “Radio Navigation”, chapters included in PPL(A) LO requirements.
3. SSFA Lecture presentations 062 Radio Navigation

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

BASIC RADIO PROPAGATION THEORY

- Antennas
- Characteristics
- Wave propagation
- Propagation with the frequency bands

RADIO AIDS

- Ground D/F
- Principles
- Presentation and interpretation
- Range
- Errors and accuracy
- Factors affecting range and accuracy.
- NDB/ADF

Principles

- Presentation and interpretation
- Range
- Errors and accuracy
- Factors affecting range and accuracy.

TEM topics:

- Equipment errors

Progress testing:

PT RNAV Lesson 01 performed at lessons end.

Interconnection:

- Instrument set up and utilization

Mandatory student preparation:

- Flygcert CBT RNAV chapters 1-3

NAVIGATION – RADIO NAVIGATION – Lesson 1

Learning Objectives

062 00 00 00 RADIO NAVIGATION
062 01 00 00 BASIC RADIO PROPAGATION THEORY
062 01 02 00 Antennas
062 01 02 01 Characteristics
062 01 03 00 Wave propagation
062 01 03 04 Propagation with the frequency bands
062 02 00 00 RADIO AIDS
062 02 01 00 Ground D/F
062 02 01 01 Principles
062 02 01 02 Presentation and interpretation
062 02 01 03 Range
062 02 01 04 Errors and accuracy
062 02 01 05 Factors affecting range and accuracy.
062 02 02 00 NDB/ADF
062 02 02 01 Principles
062 02 02 02 Presentation and interpretation
062 02 02 03 Range
062 02 02 04 Errors and accuracy
062 02 02 05 Factors affecting range and accuracy.

NAVIGATION – RADIO NAVIGATION – Lesson 2

Overview

062 02 03 00 VOR

062 02 04 00 DME

062 03 00 00 RADAR

062 06 00 00 GLOBAL NAVIGATION SATELLITE SYSTEMS

Repetition

School Test

Preparation for Exam

Ref.:

1. SSFA LO PPL(A) 062 Radio Navigation
2. Oxford or Jeppesen “Radio Navigation”, chapters included in PPL(A) LO requirements.
3. SSFA Lecture presentations 062 Radio Navigation

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

VOR

- Principles
- Presentation and interpretation
- Range
- Errors and accuracy
- Factors affecting range and accuracy.

DME

- Principles
- Presentation and interpretation
- Range
- Errors and accuracy
- Factors affecting range and accuracy

RADAR

- Ground Radar
- Principles
- Presentation and interpretation
- Secondary Surveillance Radar and transponder
- Principles
- Modes and codes
- Presentation and interpretation

GLOBAL NAVIGATION SATELLITE SYSTEMS

- GPS/GLONASS/GALILEO
- Principles
- Operation
- Error and accuracy

TEM topics:

- Equipment errors

Progress testing:

PT RNAV Lesson 02 performed at lessons end.

Interconnection:

- Instrument set up and utilization

Mandatory student preparation:

- Flygcert CBT RNAV chapters 4-6

NAVIGATION – RADIO NAVIGATION – Lesson 2

Learning Objectives

062 02 03 00 VOR
062 02 03 01 Principles
062 02 03 02 Presentation and interpretation
062 02 03 03 Range
062 02 03 04 Errors and accuracy
062 02 03 05 Factors affecting range and accuracy.
062 02 04 00 DME
062 02 04 01 Principles
062 02 04 02 Presentation and interpretation
062 02 04 03 Range
062 02 04 04 Errors and accuracy
062 02 04 05 Factors affecting range and accuracy
062 03 00 00 RADAR
062 03 02 00 Ground Radar
062 03 02 01 Principles
062 03 02 02 Presentation and interpretation
062 03 04 00 Secondary Surveillance Radar and transponder
062 03 04 01 Principles
062 03 04 02 Modes and codes
062 03 04 03 Presentation and interpretation
062 06 00 00 GLOBAL NAVIGATION SATELLITE SYSTEMS
062 06 01 00 GPS/GLONASS/GALILEO
062 06 01 01 Principles
062 06 01 02 Operation
062 06 01 03 Error and accuracy

Operational Procedures – Lesson 1

Overview

070 00 00 00 OPERATIONAL PROCEDURES

071 01 00 00 Basic Requirements

Ref.:

1. SSFA LO PPL(A) 070 Operational Procedures
2. KSAB PPL Operativa Procedurer
3. Oxford or Jeppesen “Operational Procedures”, chapters included in PPL(A) LO requirements.
4. SSFA Lecture presentations 070 Operational Procedures
5. ICAO Annex 6, 12 and 13

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

Basic requirements

- Operation of aircraft – ICAO Annex 6, General requirements
- Definitions
- Applicability
- Reporting

SPECIAL OPERATIONAL PROCEDURES AND HAZARDS (GENERAL ASPECTS)

- Runway incursion
- Noise abatement
- Noise abatement procedures
- Impact in flight procedures (departure, cruise, approach)

TEM topics:

- Airspace, runway incursion

Progress testing:

PT OPP Lesson 01 performed at lessons end.

Interconnection:

- Flight procedures

Mandatory student preparation:

- Flygcert CBT OPP chapters 1-5

Operational Procedures – Lesson 1

Learning Objectives

070 00 00 00 OPERATIONAL PROCEDURES
071 01 00 00 Basic requirements
071 01 01 00 Operation of aircraft – ICAO Annex 6, General requirements
071 01 01 01 Definitions
071 01 01 02 Applicability
071 01 02 00 Reporting
071 02 00 00 SPECIAL OPERATIONAL PROCEDURES AND HAZARDS (GENERAL ASPECTS)
071 02 03 00 Runway incursion
071 02 04 00 Noise abatement
071 02 04 01 Noise abatement procedures
071 02 04 02 Impact in flight procedures (departure, cruise, approach)

Operational Procedures – Lesson 2

Overview

071 02 00 00 SPECIAL OPERATIONAL PROCEDURES AND HAZARDS (GENERAL ASPECTS)

Repetition

School Test

Preparation for Exam

Ref.:

1. SSFA LO PPL(A) 070 Operational Procedures
2. KSAB PPL Operativa Procedurer
3. Oxford or Jeppesen “Operational Procedures”, chapters included in PPL(A) LO requirements.
4. SSFA Lecture presentations 070 Operational Procedures
5. ICAO Annex 6, 12 and 13

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

Fire/smoke

- Carburetor fire
- Engine fire
- Fire in the cabin, cockpit, (choice of extinguishing agents according to fire classification, use of the extinguishers)
- Smoke in the cockpit and cabin (effects and action to be taken)
- Windshear and microburst
- Effects and recognition during departure and approach
- Actions to avoid and actions taken during encounter.

Wake turbulence

- Cause
- List of relevant parameters
- Actions taken when crossing traffic, during takeoff and landing.

Emergency and precautionary landings

- Definition
- Cause
- Passenger information
- Action after landing
- Evacuation

Contaminated runways

- Deteriorating track conditions
- Estimated surface friction, friction coefficient

TEM topics:

- Fire, smoke, windshear, microburst

Progress testing:

PT OPP Lesson 02 performed at lessons end.

Interconnection:

- Actions taken when crossing traffic

Mandatory student preparation:

- Flygcert CBT OPP chapters 7-10

Operational Procedures – Lesson 2

Learning Objectives

071 02 05 00 Fire/smoke
071 02 05 01 Carburetor fire
071 02 05 02 Engine fire
071 02 05 03 Fire in the cabin, cockpit, (choice of extinguishing agents according to fire classification, use of the extinguishers)
071 02 05 04 Smoke in the cockpit and cabin (effects and action to be taken)
071 02 07 00 Windshear and microburst
071 02 07 01 Effects and recognition during departure and approach
071 02 07 02 Actions to avoid and actions taken during encounter.
071 02 08 00 Wake turbulence
071 02 08 01 Cause
071 02 08 02 List of relevant parameters
071 02 08 03 Actions taken when crossing traffic, during takeoff and landing.
071 02 10 00 Emergency and precautionary landings
071 02 10 01 Definition
071 02 10 02 Cause
071 02 10 03 Passenger information
071 02 10 04 Action after landing
071 02 10 05 Evacuation
071 02 13 00 Contaminated runways
071 02 13 01 Deteriorating track conditions
071 02 13 02 Estimated surface friction, friction coefficient

PRINCIPLES OF FLIGHT – AEROPLANE – Lesson 1

Overview

080 00 00 00 PRINCIPLES OF FLIGHT
081 00 00 00 PRINCIPLES OF FLIGHT – AEROPLANE
081 01 00 00 SUBSONIC AERODYNAMICS

Ref.:

4. SSFA LO PPL(A) 081 Principles of flight
5. KSAB PPL Flygningens Grundprinciper
6. Oxford or Jeppesen “PRINCIPLES OF FLIGHT”, chapters included in PPL(A) LO requirements.
7. SSFA Lecture presentations 081 Principles of flight

Setting:

Classroom

Time allocated:

Instructor led study: 4 hrs.

Lesson content

SUBSONIC AERODYNAMICS

- Basics concepts, laws and definitions
- Laws and definitions
- Basics about airflow
- streamline, two-dimensional airflow, three-dimensional airflow

Aerodynamic forces on surfaces

- resulting air force, lift, drag, angle of attack

Shape of an airfoil section:

- thickness to chord ratio, chord line, camber line, camber

The wing shape:

- aspect ratio, root chord, tip chord, tapered wings, wing planform

The two-dimensional airflow about an airfoil

- Streamline pattern
- Stagnation point
- Pressure distribution
- Centre of pressure
- impact of angle of attack on center of pressure

Flow separation at high angles of attack

- The Lift vs angle of attack graph
- The coefficients
- The lift coefficient C_L
- The drag coefficient C_d
- The three-dimensional airflow round a wing and a fuselage

Streamline pattern:

- span wise flow, tip vortices, upwash and downwash, wake turbulence

Induced drag

Total Drag

The parasite drags.

- form drags, interference drags, friction drags
- The parasite drags and speed.
- The induced drag and speed
- The total drag
- The ground effect
- Effect on takeoff and landing characteristics of an airplane

The stall

- Flow separation at increasing angles of attack

- The stall speed
- The initial stall
- Stall warning
- Special phenomena of stall

CL augmentation

- Trailing edge flaps
- Leading edge devices

The boundary layer

- Characteristic

Special circumstances (what has negative impact on aerodynamic)**Ice and other contamination****TEM topics:**

- Stall, spin

Progress testing:

PT POF Lesson 01 performed at lessons end.

Interconnection:

- Slow flight, stall, incipient spin

Mandatory student preparation:

- Flygcert CBT POF chapters 1-3

PRINCIPLES OF FLIGHT – AEROPLANE – Lesson 1

Learning Objectives

080 00 00 00 PRINCIPLES OF FLIGHT
 081 00 00 00 PRINCIPLES OF FLIGHT – AEROPLANE
 081 01 00 00 SUBSONIC AERODYNAMICS
 081 01 01 00 Basics concepts, laws and definitions
 081 01 01 01 Laws and definitions
 081 01 01 02 Basics about airflow
 - streamline, two-dimensional airflow, three-dimensional airflow
 081 01 01 03 Aerodynamic forces on surfaces
 - resulting air force, lift, drag, angle of attack
 081 01 01 04 Shape of an airfoil section:
 - thickness to chord ratio, chord line, camber line, camber
 081 01 01 05 The wing shape:
 - aspect ratio, root chord, tip chord, tapered wings, wing planform
 081 01 02 00 The two-dimensional airflow about an airfoil
 081 01 02 01 Streamline pattern
 081 01 02 02 Stagnation point
 081 01 02 03 Pressure distribution
 081 01 02 04 Centre of pressure
 081 01 02 05 impact of angle of attack on center of pressure
 081 01 02 06 Flow separation at high angles of attack
 081 01 02 07 The Lift vs angle of attack graph
 081 01 03 00 The coefficients
 081 01 03 01 The lift coefficient CL
 081 01 03 02 The drag coefficient Cd
 081 01 04 00 The three-dimensional airflow round a wing and a fuselage
 081 01 04 01 Streamline pattern:
 - span wise flow, tip vortices, upwash and downwash, wake turbulence
 081 01 04 02 Induced drag
 081 01 05 00 Total Drag
 081 01 05 01 The parasite drags.
 – form drags, interference drags, friction drags
 081 01 05 02 The parasite drags and speed.
 081 01 05 03 The induced drag and speed
 081 01 05 05 The total drag
 081 01 06 00 The ground effect
 081 01 06 04 Effect on takeoff and landing characteristics of an aeroplane
 081 01 08 00 The stall
 081 01 08 01 Flow separation at increasing angles of attack
 081 01 08 02 The stall speed
 081 01 08 03 The initial stall
 081 01 08 04 Stall warning
 081 01 08 05 Special phenomena of stall
 081 01 09 00 CL augmentation
 081 01 09 01 Trailing edge flaps
 081 01 09 02 Leading edge devices
 081 01 11 00 The boundary layer
 081 01 11 01 Characteristic
 081 01 12 00 Special circumstances (what has negative impact on aerodynamic)
 081 01 12 01 Ice and other contamination

PRINCIPLES OF FLIGHT – AEROPLANE – Lesson 2

Overview

081 04 00 00 STABILITY

Ref.:

1. SSFA LO PPL(A) 081 Principles of flight
2. KSAB PPL Flygningens Grundprinciper
3. Oxford or Jeppesen “PRINCIPLES OF FLIGHT”, chapters included in PPL(A) LO requirements.
4. SSFA Lecture presentations 081 Principles of flight

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

Static and Dynamic stability

- Concepts and definitions

Static stability

- Forces equilibrium
- Moments equilibrium
- Static and dynamic longitudinal stability
- Method to achieve equilibrium.
- Static longitudinal stability
- Neutral point
- Location of center of gravity
- Dynamic lateral/directional stability
- Spiral dive, corrective actions

TEM topics:

- Equipment errors

Progress testing:

PT POF Lesson 02 performed at lessons end.

Interconnection:

- Coordinated flight

Mandatory student preparation:

- Flygcert CBT POF chapters 4-6

PRINCIPLES OF FLIGHT – AEROPLANE – Lesson 2

Learning Objectives

081 04 00 00 STABILITY
081 04 01 00 Static and Dynamic stability
081 04 01 01 Concepts and definitions
081 04 01 02 Static stability
081 04 01 03 Forces equilibrium
081 04 01 04 Moments equilibrium
081 04 03 00 Static and dynamic longitudinal stability
081 04 03 01 Method to achieve equilibrium.
081 04 03 02 Static longitudinal stability
081 04 03 03 Neutral point
081 04 03 05 Location of center of gravity
081 04 06 00 Dynamic lateral/directional stability
081 04 06 02 Spiral dive, corrective actions

PRINCIPLES OF FLIGHT – AEROPLANE – Lesson 3

Overview

081 05 00 00 CONTROL

081 06 00 00 LIMITATIONS

Ref.:

5. SSFA LO PPL(A) 081 Principles of flight
6. KSAB PPL Flygningens Grundprinciper
7. Oxford or Jeppesen “PRINCIPLES OF FLIGHT”, chapters included in PPL(A) LO requirements.
8. SSFA Lecture presentations 081 Principles of flight

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

CONTROL

General

- Basics, the three planes and three axes
- Angle of attack change
- Pitch control
- Elevator
- Downwash effects
- Location of center of gravity
- Yaw control
- Rudder
- Roll control

Ailerons

- Adverse yaw
- Means to avoid adverse yaw.
- Means to reduce control forces.
- Aerodynamic balance

Trimming

- Reasons to trim
- Trim tabs

LIMITATIONS

Operating limitations

- Flutter
- VFE
- VNO, VNE
- Maneuvering envelope
- Load factor
- Factors affecting load factor
- Gust envelope
- Gust load diagram

TEM topics:

- Maneuvering envelope, load factors

Progress testing:

PT POF Lesson 01 performed at lessons end.

Interconnection:

- Slip, skid, sideslip

Mandatory student preparation:

- Flygcert CBT POF chapters 7-9

PRINCIPLES OF FLIGHT – AEROPLANE – Lesson 3

Learning Objectives

081 05 00 00 CONTROL
081 05 01 00 General
081 05 01 01 Basics, the three planes and three axes
081 05 01 03 Angle of attack change
081 05 02 00 Pitch control
081 05 02 01 Elevator
081 05 02 02 Downwash effects
081 05 02 04 Location of center of gravity
081 05 03 00 Yaw control
081 05 03 01 Rudder
081 05 04 00 Roll control
081 05 04 01 Ailerons
081 05 04 04 Adverse yaw
081 05 04 05 Means to avoid adverse yaw.
081 05 06 00 Means to reduce control forces.
081 05 06 01 Aerodynamic balance
081 05 08 00 Trimming
081 05 08 01 Reasons to trim
081 05 08 02 Trim tabs
081 06 00 00 LIMITATIONS
081 06 01 00 Operating limitations
081 06 01 01 Flutter
081 06 01 03 VFE
081 06 01 04 VNO, VNE
081 06 02 00 Maneuvering envelope
081 06 02 01 Load factor
081 06 02 02 Factors affecting load factor
081 06 03 00 Gust envelope
081 06 03 01 Gust load diagram

PRINCIPLES OF FLIGHT – AEROPLANE – Lesson 4

Overview

081 07 00 00 PROPELLERS

081 08 00 00 FLIGHT MECHANICS

Repetition

School Test

Preparation for Exam

Ref.:

1. SSFA LO PPL(A) 081 Principles of flight
2. KSAB PPL Flygningens Grundprinciper
3. Oxford or Jeppesen “PRINCIPLES OF FLIGHT”, chapters included in PPL(A) LO requirements.
4. SSFA Lecture presentations 081 Principles of flight

Setting:

Classroom

Time allocated:

Instructor led study: 3 hrs.

Lesson content

PROPELLERS

- Conversion of engine torque to thrust.
- Meaning of pitch
- Blade twist
- Effects of ice on propeller
- Engine failure or engine stop
- Drag caused by propeller.
- Secondary propeller effects
- Torque reaction
- Slipstream effect
- P-Factor
- Forces in different phases of flight
- Level flight
- Climb
- Descend and Gliding
- Turning

TEM topics:

- Equipment errors

Progress testing:

PT POF Lesson 04 performed at lessons end.

Interconnection:

- Instrument set up and utilization

Mandatory student preparation:

- Flygcert CBT POF chapters 10-11

PRINCIPLES OF FLIGHT – AEROPLANE – Lesson 4

Learning Objectives

081 07 00 00 PROPELLERS
081 07 01 00 Conversion of engine torque to thrust.
081 07 01 01 Meaning of pitch
081 07 01 02 Blade twist
081 07 01 05 Effects of ice on propeller
081 07 02 00 Engine failure or engine stop
081 07 02 01 Drag caused by propeller.
081 07 04 00 Secondary propeller effects
081 07 04 01 Torque reaction
081 07 04 03 Slipstream effect
081 07 04 04 P-Factor
081 08 00 00 Forces in different phases of flight
081 08 01 01 Level flight
081 08 01 02 Climb
081 08 01 03 Descend and Gliding
081 08 01 05 Turning

Communication – Lesson 1

Overview

090 00 00 00 COMMUNICATIONS
091 00 00 00 VFR COMMUNICATIONS
091 01 00 00 DEFINITIONS
091 02 00 00 GENERAL OPERATING PROCEDURES
091 03 00 00 RELEVANT WEATHER INFORMATION TERMS (VFR)
091 04 00 00 ACTION REQUIRED TO BE TAKEN IN CASE OF COMMUNICATION FAILURE
091 05 00 00 DISTRESS AND URGENCY PROCEDURES
091 06 00 00 GENERAL PRINCIPLES OF VHF PROPAGATION

Repetition
School Test
Preparation for Exam

Ref.:

1. SSFA LO PPL(A) 090 Communication
2. KSAB PPL Flygradiotelefon
3. Oxford or Jeppesen “Communication”, chapters included in PPL(A) LO requirements.
4. SSFA Lecture presentations 090 Communication
5. ICAO doc 4444

Introduction

Setting:
Classroom

Time allocated:
Instructor led study: 3 hrs.

Lesson content

FLIGHT RADIO TELEPHONY

VFR communications

definitions

- terms and abbreviation
- air traffic services abbreviations
- Q-code groups commonly used in communications.
- categories of messages

general operating procedures

- transmission of letters
- transmission of numbers (including level information)
- transmission of time
- transmission technique
- standard words and phrases (relevant rtf phraseology included)
- radiotelephony call signs for aeronautical stations including use of abbreviated call signs.

changing of frequency.

- radio test procedures
- read back and acknowledgement requirements.

relevant weather information terms (VFR)

- aerodrome weather
- in-flight weather information

communication failure

- action required to be taken in case of communication failure.
- distress procedures

- distress
- urgency procedures

general principles of vhf propagation

- VHF frequency

TEM topics:

- Loss of com procedures

Progress testing:

PT COM Lesson 01 performed at lessons end.

Interconnection:

- Use of coms, set up, double check, trouble shooting

Mandatory student preparation:

- Flygcert CBT COM chapters 1-5

Communication – Lesson 1

090 00 00 00 flight radio telephony
091 00 00 00 VFR communications
091 01 00 00 definitions
091 01 01 00 terms and abbreviation
091 01 02 00 air traffic services abbreviations
091 01 03 00 Q-code groups commonly used in communications.
091 01 04 00 categories of messages
091 02 00 00 general operating procedures
091 02 01 00 transmission of letters
091 02 02 00 transmission of numbers (including level information)
091 02 03 00 transmission of time
091 02 04 00 transmission technique
091 02 05 00 standard words and phrases (relevant rtf phraseology included)
091 02 06 00 radiotelephony call signs for aeronautical stations including use of abbreviated call signs.

091 02 07 00 changing of frequency.
091 02 08 00 radio test procedures
091 02 09 00 read back and acknowledgement requirements.
091 03 00 00 relevant weather information terms (VFR)
091 03 01 00 aerodrome weather
091 03 02 00 in-flight weather information
091 04 00 00 communication failure
091 04 01 00 action required to be taken in case of communication failure.
091 05 00 00 distress procedures
091 05 01 00 distress
091 05 02 00 urgency procedures
091 06 00 00 general principles of vhf propagation
091 06 01 00 VHF frequency